METHOTREXATE

ONCE WEEKLY for RA/psoriasis

Fatal daily-dosing errors

CHEMICAL NAME (2S)-2-[(4-{[(2,4-diaminopteridin-6-yl)methyl](methyl)amino}benzoyl)amino]pentanedioic acid

CLASS Antimetabolite (folate antagonist) — DMARD and cytotoxic

TRADE NAMES Maxtrex · Metoject (SC) · Trexall · Otrexup

THERAPEUTIC USES / INDICATIONS

- Rheumatoid arthritis (first-line DMARD), psoriatic arthritis, juvenile idiopathic arthritis.
- Severe psoriasis; Crohn's disease.
- Cancers: ALL, lymphoma, osteosarcoma, choriocarcinoma, breast cancer (high doses).
- Ectopic pregnancy (single IM dose); medical abortion (some protocols).

ADMINISTRATION

- **PO** or **SC** ONCE A WEEK on the same day for inflammatory disease — 2.5 mg tablets only (avoid 10 mg strength confusion).
- Folic acid on a different day each week.
- Cytotoxic handling precautions (gloves; pregnant staff avoid handling).
- High-dose IV oncology regimens require leucovorin rescue and urine alkalinisation.

DOSAGES

- **RA:** 7.5–10 mg once weekly, increase to 15–25 mg once weekly.
- **Folic acid:** 5 mg once weekly (not on methotrexate day) or 1 mg daily except MTX day.
- **Ectopic pregnancy:** 50 mg/m² IM single dose.
- **Oncology:** protocol-specific (up to grams/m² with leucovorin rescue).
- Reduce in renal impairment; avoid if eGFR <30.

PHARMACOKINETICS

- **Onset:** RA benefit in 4–8 weeks.
- Bioavailability decreases with higher oral doses (SC better).
- Excreted mainly unchanged in urine (accumulates in renal failure, third-space fluids — ascites, effusions). **t½:** 3–10 h (low dose).

PHARMACODYNAMICS

- Inhibits **dihydrofolate reductase** → ↓ tetrahydrofolate → blocks DNA synthesis in rapidly dividing cells (cytotoxic).
- Low dose: ↑ adenosine release → anti-inflammatory/immunomodulatory effect.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, mouth ulcers, fatigue (day after dose).
- Hair thinning.
- Raised liver enzymes.

ADVERSE EFFECTS (SERIOUS)

- Bone-marrow suppression (pancytopenia) — fatal with daily dosing errors or interactions.
- Hepatotoxicity, fibrosis, cirrhosis.
- Pneumonitis (cough, breathlessness) and pulmonary fibrosis.
- Nephrotoxicity (high dose); severe infections; teratogenic (abortifacient).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (stop 3–6 months before conception in men and women), breastfeeding.
- Significant renal or hepatic impairment; alcohol misuse.
- Active infection, immunodeficiency; pre-existing blood dyscrasias.
- Live vaccines.

KEY INTERACTIONS

- Trimethoprim, co-trimoxazole: severe marrow suppression; avoid.
- **NSAIDs, aspirin (high dose), PPIs (high-dose MTX), penicillins, probenecid:** ↓ excretion → toxicity.
- **Alcohol, leflunomide:** ↑ hepatotoxicity.
- **Live vaccines:** contraindicated.

NURSING MANAGEMENT

- Verify weekly day of dosing on every prescription and chart.
- Baseline FBC, U&E, LFTs, chest X-ray; FBC/LFT/U&E every 2 weeks until stable for 6 weeks, then monthly → 3-monthly.
- Ask about sore throat, mouth ulcers, bruising, cough, breathlessness.
- Cytotoxic waste disposal; gloves when handling.
- Contraception counselling.

PATIENT EDUCATION

- Take ONCE A WEEK on the same day; never daily.
- Take folic acid on a different day as prescribed.
- Report sore throat, fever, mouth ulcers, bruising, new cough or breathlessness.
- Avoid alcohol (or keep well within limits); avoid pregnancy; no live vaccines.

KEY POINTS

- DHFR inhibitor — DMARD & cytotoxic.
- Marrow, liver, lung toxicity; regular bloods.
- RA: once WEEKLY 7.5–25 mg + folic acid.
- Never with trimethoprim; teratogenic.

SOURCES CHECKED **[840]** Methotrexate 10 mg Tablets (UK SmPC) **[841]** Methotrexate 100 mg/ml Injection (UK SmPC) **[842]** Methotrexate, intra-arterial, IM, intrathecal, IV (US label) **[843]** Methotrexate, oral (US label)

Full references p. 432–457